



Guidelines

Empiric Antibiotic Treatment for Intra-Abdominal Infection

Scope

Site	Service/Department/Unit	Disciplines
SCGH and OPH (i.e. SCGOPHCG)	Organisation Wide	Medical, Nursing, Pharmacy

Scope

This guideline provides empiric antimicrobial therapy recommendations for inpatients with intra-abdominal infection. It serves as a guide only and infectious diseases advice should be sought in instances where prolonged antimicrobials are required, signs of septic shock are evident or microbiological investigations isolate a causative organism.

Introduction

Optimal empirical antibiotic treatment of intra-abdominal infections involves the use of appropriate antibiotics with the narrowest spectrum required for effective treatment administered for the shortest duration to achieve clinical resolution. This reduces the risk of adverse effects including infection with *Clostridioides difficile* and multi-resistant organisms.¹

Most intra-abdominal infections are polymicrobial and are often caused by intestinal bacteria. Enterobacteriaceae and streptococci are common and empirical antibiotic choice should include antibiotics with activity against these organisms.

Additional activity against enteric anaerobic bacteria is recommended if the distal small bowel, appendix, or colon are involved or for more proximal gastrointestinal perforations in the presence of obstruction or paralytic ileus.

Assessment

- Assess the patient for associated systemic sepsis or septic shock.

Please refer to SCGOPHCG Adult Sepsis Pathway for more detail <https://scgophcg-healthpoint.hdwa.health.wa.gov.au/directory/Corporate/SQP/NationalStandards/Documents/Adult%20Sepsis%20Pathway.pdf>

Investigations

- Obtain samples from site of infection for culture and susceptibility testing such as abscess pus, intra-operative specimens and blood cultures.
- Please refer to specimen collection guideline:
https://healthpoint.hdwa.health.wa.gov.au/policies/Policies/NMAHS/SCGH/SCGH.IPCM.Specimen_Collection_for_Microbiological_Examination_guideline.pdf
 - Infected fluid, pus or tissue samples are preferable to swabs.
- Avoid using drain samples for directing therapy as these are often contaminated and not an accurate representation of the true pathogens present
- Treatment choice should always be guided by positive culture results if available (directed antibiotic therapy).

Management

- Refer to **Table 1** below for guidance on empiric antimicrobial choice and duration.
- Intravenous antibiotics should be started initially, with oral switch once criteria have been met (see below)
- Criteria for oral switch include¹
 - Patient displaying signs of clinical improvement (afebrile >24 hours, down trending inflammatory markers, no unexplained hypotension or tachycardia)
 - Uncomplicated infection
 - Able to tolerate oral or enteral therapy
 - Adequate oral absorption
 - e.g. patient does not have a high stoma output, high nasogastric aspirates, dysphagia or bowel obstruction/ileus
- In the absence of complications, the duration of antimicrobial therapy should generally be no more than five (5) to seven (7) days total (total intravenous and oral therapy).
- Patients who continue to exhibit clinical evidence of infection should be evaluated for residual infection (undrained collection/abscess or other remaining source), resistant organisms, non-infectious causes of inflammation and other possible reasons of treatment failure (e.g. septic thrombophlebitis). Seek advice from Infectious Diseases (ID) or a Clinical Microbiologist.

- Patients with non-severe penicillin allergy should be referred to the ID Penicillin Allergy Delabelling team via a HAPI referral: [Low Risk Penicillin Delabelling Service \(health.wa.gov.au\)](https://www.health.wa.gov.au/low-risk-penicillin-delabelling-service)



Table 1: Initial Empirical Antibiotic Therapy for Intra-Abdominal Infections¹

(oral options serve as a guide only – if positive microbiology available then the oral choice should be guided by any susceptibility results)

Indication	Route	First Line	Non-Severe Penicillin Allergy	Severe Penicillin Allergy	Total Duration (IV/PO) + Notes
Acute Uncomplicated Appendicitis Inflammation of the appendiceal wall without any perforation or abscess formation	IV	Amoxicillin/clavulanic acid 1.2g 6 hourly (Reduce to 12 hourly if CrCL <30mL/min)	Ceftriaxone 2g daily PLUS Metronidazole IV 500mg 12 hourly (change to PO formulation 400mg 12 hourly when able to tolerate)	Clindamycin IV 600mg 8 hourly PLUS Ciprofloxacin IV 400mg 12 hourly	Discontinue antibiotics immediately after appendicectomy (if no complications)
	No oral step-down for uncomplicated appendicitis. Antibiotics should be discontinued post-operatively.				
Acute Complicated Appendicitis Involves either perforation of the appendix or presence of an appendiceal abscess	IV	Amoxicillin/clavulanic acid 1.2g 6 hourly (Reduce to 12 hourly if CrCL <30mL/min)	Ceftriaxone 2g daily PLUS Metronidazole IV 500 mg 12 hourly (change to PO formulation 400mg 12 hourly when able to tolerate)	Clindamycin IV 600mg 8 hourly PLUS Ciprofloxacin IV 400mg 12 hourly	Discontinue antibiotics (IV + PO) 4 days after source control achieved
	Oral switch	Amoxicillin/clavulanic acid 875/125mg 12 hourly	Cefuroxime 500mg 12 hourly PLUS Metronidazole 400mg 12 hourly	Trimethoprim/sulfamethoxazole 160/800mg 12 hourly PLUS Metronidazole 400mg 12 hourly	



Indication	Route	First Line	Non-Severe Penicillin Allergy	Severe Penicillin Allergy	Total Duration (IV/PO) + Notes
Acute Calculous Cholecystitis ○ Acute inflammation of the gallbladder wall due to obstruction of the cystic duct by gallstones	IV	Amoxicillin/clavulanic acid 1.2g 6 hourly (Reduce to 12 hourly if CrCL <30mL/min)	Ceftriaxone 2g daily	Clindamycin IV 600mg 8 hourly PLUS Ciprofloxacin IV 400mg 12 hourly	Discontinue antibiotics immediately after cholecystectomy
	No oral step-down unless residual infection suspected	Amoxicillin/clavulanic acid 875/125mg 12 hourly	Trimethoprim/sulfamethoxazole 160/800mg 12 hourly		Discontinue antibiotics post-operatively as the source of infection has been removed. If residual infection is suspected OR awaiting cholecystectomy give a maximum of 7days (IV + PO post-operatively)
Acute Acalculous Cholecystitis ○ Acute inflammation of the gallbladder wall in the absence of gallstones, and generally associated with dysfunction or hypokinesia of gallbladder emptying	IV	Amoxicillin/clavulanic acid 1.2g 6 hourly (Reduce to 12 hourly if CrCL <30mL/min)	Ceftriaxone 2g daily	Clindamycin IV 600mg 8 hourly PLUS Ciprofloxacin IV 400mg 12 hourly	Continue antibiotics for up to 4 days after source control achieved (e.g. cholecystectomy or cholecystostomy insertion)
	Oral switch: call for advice if resistant pathogen identified	Amoxicillin/clavulanic acid 875/125mg 12 hourly	Trimethoprim/sulfamethoxazole 160/800mg 12 hourly PLUS Metronidazole 400mg 12 hourly		If no surgical intervention performed, contact ID or HAPI for guidance



Indication	Route	First Line	Non-Severe Penicillin Allergy	Severe Penicillin Allergy	Total Duration (IV/PO) + Notes
Acute (ascending) Cholangitis A life-threatening condition involving infection of the biliary tree, caused by an ascending bacterial infection from the duodenum	IV	Piperacillin/tazobactam 4.5g 8 hourly (CrCL >30mL/min) If CrCL <20mL/min, give 12 hourly	Ceftriaxone 2g daily If chronic biliary obstruction, add Metronidazole 500mg 12 hourly (change to PO formulation 400mg 12 hourly when able to tolerate)	Clindamycin IV 600mg 8 hourly PLUS Ciprofloxacin IV 400mg 12 hourly	If biliary drainage is performed, continue antibiotics (IV + PO) for a total of 4 days only post-operatively If no biliary drainage performed, the total duration of antibiotics is 7 days (IV + PO)
	Oral switch	Amoxicillin/clavulanic acid 875/125mg 12 hourly If no biliary drainage performed, use Amoxicillin/clavulanic acid 875/125mg <u>8 hourly</u>	Trimethoprim/sulfamethoxazole 160/800mg 12 hourly If chronic obstruction, add Metronidazole 400mg 12 hourly		
Uncomplicated (non-severe) Diverticulitis - Inflammation of small, bulging pouches (diverticula) within the wall of the large intestine. - Uncomplicated cases do not involve any perforation, peritonitis, sepsis or large abscess	IV	Uncomplicated diverticulitis may not require intravenous antibiotics. If IV required follow complicated diverticulitis regimens.			
	Oral options	Amoxicillin/clavulanic acid 875/125mg 12 hourly	Cefuroxime 500mg 12 hourly PLUS Metronidazole 400mg 12 hourly	Trimethoprim/sulfamethoxazole 160/800mg 12 hourly PLUS Metronidazole 400mg 12 hourly	Total duration of 5-days only



Indication	Route	First Line	Non-Severe Penicillin Allergy	Severe Penicillin Allergy	Total Duration (IV/PO) + Notes
Complicated Diverticulitis Complicated disease includes cases where there is a bacteraemia, perforation, peritonitis, sepsis or an abscess larger than 5cm	IV	Amoxicillin/clavulanic acid 1.2g 6 hourly (12 hourly if CrCL <30mL/min)	Ceftriaxone 2g daily PLUS Metronidazole 500 mg 12 hourly (change to PO formulation 400mg 12 hourly when able to tolerate)	Clindamycin IV 600mg 8 hourly PLUS Ciprofloxacin IV 400mg 12 hourly NB. If prompt source control unlikely then consider adding metronidazole to regimen	If surgical intervention / drainage performed, total duration is 4 days (IV + PO) after procedure/source control If no surgical intervention the total duration is 7-10 days (IV + PO)
	Oral switch	Amoxicillin/clavulanic acid 875/125mg 12 hourly	Cefuroxime 500mg 12 hourly PLUS Metronidazole 400mg 12 hourly	Trimethoprim/sulfamethoxazole 160/800mg 12 hourly PLUS Metronidazole 400mg 12 hourly	Switch when clinically improving, haemodynamically stable and tolerating oral therapy
Peritonitis due to perforated viscus Perforation of the GI tract with contamination of the peritoneum	IV	Piperacillin/tazobactam 4.5g 8 hourly (Reduce to 12 hourly if CrCL <20mL/min)	Ceftriaxone 2g daily PLUS Metronidazole 500mg 12 hourly (change to PO formulation 400mg 12 hourly when able to tolerate)	Clindamycin IV 600mg 8 hourly PLUS Ciprofloxacin IV 400mg 12 hourly	Continue for 4 days (IV+ PO) after adequate source control achieved. If patient critically unwell after source control then duration may be extended up to 7 days.
	Oral switch	Amoxicillin/clavulanic acid 875/125mg 12 hourly	Trimethoprim/sulfamethoxazole 160/800mg 12 hourly PLUS Metronidazole 400mg 12 hourly		If residual or undrained abscess/ collection, a longer duration may be required (up to 4-6 weeks). Contact ID for advice.



Indication	Route	First Line	Non-Severe Penicillin Allergy	Severe Penicillin Allergy	Total Duration (IV/PO) + Notes
Acute Infected Pancreatitis Inflammation of the pancreas with resulting development of pancreatic fluid collections that are suspected or proven to be infected	IV	Piperacillin/tazobactam 4.5g 6 hourly if CrCL 20-40mL/min, 8 hourly If CrCL <20mL/min, give 12 hourly	Ceftriaxone 2g daily PLUS Metronidazole 500mg 12 hourly (change to PO formulation 400mg 12 hourly when able to tolerate) PLUS Vancomycin - 15mg/kg of ABW 12 hourly (CrCl > 60ml/min) If CrCl < 60ml/min, seek advice from Clinical Pharmacist for dosing guidance	Contact ID for advice	If adequate surgical source control is achieved then cease therapy after 5 to 7 days. If infection not resolved within 7 to 10 days then seek ID or HAPI advice for further guidance

Antimicrobial Restrictions

- Indication/s for antibiotic therapy should be clearly documented on the medication chart and within the inpatient notes.
- If there is a positive microbiology result, or if the patient is displaying signs of sepsis or not responding to antibiotics then ID or CM input should be sought asap (either via phone advice or HAPI referral)
 - For detail into antimicrobial prescribing restrictions please see Table 2 below

Table 2: Antimicrobial Prescribing Restrictions and HAPI Referral Requirements

Antimicrobial	Prescribing Restrictions
a. Piperacillin-tazobactam (IV) b. Ceftriaxone (IV)	If prescribed in accordance with these guidelines, can be initiated without ID/Micro approval and continued for up to 72 hours. If therapy continues beyond 72 hours then a HAPI referral must be submitted.
a. Amoxicillin-clavulanate (IV and oral) b. Metronidazole (IV and oral) c. Sulfamethoxazole-trimethoprim (IV and oral)	If prescribed in accordance with these guidelines, can be initiated without ID/Micro approval and continued for the recommended durations as described in Table 1. If therapy continues beyond the recommended duration/s provided in Table 1 then a HAPI referral must be submitted.

Related Documents

Authorising Policy and Standard/s:

- NSQHS Standard: [Sepsis, Antimicrobial Stewardship](#)

Standards, procedures, guidelines:

- [Sepsis Management Clinical Guideline](#)

References

1. Antibiotic Expert Groups, Therapeutic Guidelines: Antibiotic. Version 16. 2019 Melbourne: Therapeutic Guidelines Limited.

Adapted from Royal Perth Hospital guideline “Empirical Antibiotic Treatment of Intra-Abdominal Infections in Hospitalised Patients Clinical Guideline Scope”. We would like to gratefully acknowledge the work of the authors and contributors documented below.

- Maryam Sherkat Masoum, Antimicrobial Stewardship Pharmacist, RPH
- Alice Hashiguchi, A/Antimicrobial Stewardship Pharmacist, RPH
- Paul Ingram, ID Consultant and Clinical Microbiologist, RPH

Authorisation & Version Control

Policy Sponsor:	SCGOPHCG Director of Clinical Services				
Policy Contact:	SCGOPHCG Medicines and Therapeutics Committee Executive Officer (Senior Pharmacist)				
Date First Issued:	12/05/2025	Last Reviewed:	7/08/2025	Review Date:	07/08/2028
Version No.	1.1	Reference #:	MMG059		
Approved by:	SCGOPHCG Antimicrobial Stewardship Committee			Date:	27/02/2025
Endorsed by:	SCGOPHCG Medicines and Therapeutics Committee			Date:	12/05/2025
Acknowledgements:	Authors: Jason Seet (Infectious Diseases Pharmacist) Dr Julie Hart (Microbiology Consultant)				
Standards Applicable	NSQHS Standards 				



APPENDIX 1

Initial Empirical Antibiotic Therapy for Intra-Abdominal Infections

(oral options serve as a guide only – if positive microbiology available then the oral choice should be guided by any susceptibility results)

Indication	Route	First Line	Non-Severe Penicillin Allergy	Severe Penicillin Allergy	Total Duration (IV/PO) + Notes
Acute Uncomplicated Appendicitis	IV	Amoxicillin/clavulanic acid 1.2g 6 hourly	Ceftriaxone 2g daily PLUS Metronidazole IV 500mg 12 hourly	Clindamycin IV 600mg 8 hourly PLUS Ciprofloxacin IV 400mg 12 hourly	Discontinue antibiotics immediately after appendectomy
	No oral step-down for uncomplicated appendicitis. Antibiotics can be discontinued post-operatively.				
Acute Complicated Appendicitis	IV	Amoxicillin/clavulanic acid 1.2g 6 hourly	Ceftriaxone 2g daily PLUS Metronidazole IV 500 mg 12 hourly	Clindamycin IV 600mg 8 hourly PLUS Ciprofloxacin IV 400mg 12 hourly	Discontinue antibiotics 4 days after source control achieved
	Oral switch	Amoxicillin/clavulanic acid 875/125mg 12 hourly	Cefuroxime 500mg 12 hourly PLUS Metronidazole 400mg 12 hourly	Trimethoprim/sulfamethoxazole 160/800mg 12 hourly PLUS Metronidazole 400mg 12 hourly	
Acute Calculous Cholecystitis	IV	Amoxicillin/clavulanic acid 1.2g 6 hourly	Ceftriaxone 2g daily	Clindamycin IV 600mg 8 hourly PLUS Ciprofloxacin IV 400mg 12 hourly	Discontinued antibiotics immediately after cholecystectomy
	Oral switch	Oral step-down only if residual infection present: Amoxicillin/clavulanic acid 875/125mg 12 hourly	Trimethoprim/sulfamethoxazole 160/800mg 12 hourly		If residual infection is suspected or awaiting cholecystectomy give a max of 7 days (IV + PO post-op)
Acute Acalculous Cholecystitis	IV	Amoxicillin/clavulanic acid 1.2g 6 hourly	Ceftriaxone 2g daily	Clindamycin IV 600mg 8 hourly PLUS Ciprofloxacin IV 400mg 12 hourly	Continue antibiotics for up to 4 days after source control achieved (e.g. cholecystectomy or cholecystostomy insertion)
	Oral switch	Amoxicillin/clavulanic acid 875/125mg 12 hourly (or directed therapy)	Trimethoprim/sulfamethoxazole 160/800mg 12 hourly PLUS Metronidazole 400mg 12 hourly		If no surgical intervention performed, contact ID or HAPI for guidance

Indication	Route	First Line	Non-Severe Penicillin Allergy	Severe Penicillin Allergy	Total Duration (IV/PO) + Notes
Acute (ascending) Cholangitis	IV	Piperacillin/tazobactam 4.5g 8 hourly	Ceftriaxone 2g daily If chronic biliary obstruction, add Metronidazole 500mg 12 hourly	Clindamycin IV 600mg 8 hourly PLUS Ciprofloxacin IV 400mg 12 hourly	If biliary drainage is performed, continue antibiotics (IV + PO) for a total of 4 days only after procedure
	Oral switch	Amoxicillin/clavulanic acid 875/125mg 12 hourly (If no drainage give 8 hourly)	Trimethoprim/sulfamethoxazole 160/800mg 12 hourly If chronic obstruction, add Metronidazole 400mg 12 hourly		If no biliary drainage performed, the total duration of antibiotics is 7-10 days (IV + PO)
Uncomplicated (non-severe) Diverticulitis	IV	Uncomplicated diverticulitis may not require intravenous antibiotics. If IV required follow complicated diverticulitis regimens.			
	Oral options	Amoxicillin/clavulanic acid 875/125mg 12 hourly	Cefuroxime 500mg 12 hourly PLUS Metronidazole 400mg 12 hourly	Trimethoprim/sulfamethoxazole 160/800mg 12 hourly PLUS Metronidazole 400mg 12 hourly	Total duration of 5-days only
Complicated Diverticulitis	IV	Amoxicillin/clavulanic acid 1.2g 6 hourly	Ceftriaxone 2g daily PLUS Metronidazole 500 mg 12 hourly	Clindamycin IV 600mg 8 hourly PLUS Ciprofloxacin IV 400mg 12 hourly Consider metronidazole if no prompt source control	If surgical intervention / drainage performed, total duration is 4 days after procedure/source control If no surgical intervention the total duration is 7-10 days (IV +PO)
	Oral switch	Amoxicillin/clavulanic acid 875/125mg 12 hourly	Cefuroxime 500mg 12 hourly PLUS Metronidazole 400mg 12 hourly	Trimethoprim/sulfamethoxazole 160/800mg 12 hourly PLUS Metronidazole 400mg 12 hourly	Switch when clinically improving, haemodynamically stable and tolerating oral therapy
Peritonitis due to perforated viscus	IV	Piperacillin/tazobactam 4.5g 8 hourly	Ceftriaxone 2g daily PLUS Metronidazole 500mg 12 hourly	Clindamycin IV 600mg 8 hourly PLUS Ciprofloxacin IV 400mg 12 hourly	Continue for 5 days (IV/PO) after adequate source control has been achieved.
	Oral switch	Amoxicillin/clavulanic acid 875/125mg 12 hourly	Trimethoprim/sulfamethoxazole 160/800mg 12 hourly PLUS Metronidazole 400mg 12 hourly		If residual or undrained abscess/collection, a longer duration may be required. Contact ID for advice.
Acute infected pancreatitis	IV	Piperacillin/tazobactam 4.5g 6 hourly	Ceftriaxone 2g daily PLUS Metronidazole 500 mg 12 hourly PLUS Vancomycin 15mg/kg of ABW 12 hourly. Seek advice if reduced CrCl	Seek ID advice	If adequate surgical source control achieved, cease after 5 to 7 days. If infection unresolved after 7 to 10 days refer to ID or HAPI



DOSAGE ADJUSTMENT GUIDE

Piperacillin-tazobactam		Amoxicillin-clavulanate			Ciprofloxacin IV		Trimethoprim-sulfamethoxazole oral	
			IV	oral				
CrCL > 40ml/min	Normal dosing	CrCL > 50ml/min	Normal dosing	Normal dosing	CrCL > 50ml/min	Normal dosing	CrCL > 50ml/min	Normal dosing
CrCL 20 to 40ml/min	4.5g 8-hourly	CrCL 30 to 50ml/min	Normal dosing	Normal dosing	CrCL 31 to 50ml/min	400mg 12-hourly	CrCL 26 to 50ml/min	160/800mg 12 hourly for 14 days then reduce to 80-400mg 12 hourly
CrCL < 20ml/min	4.5g 12-hourly	CrCL < 30ml/min	1.2g 12-hourly	500/125mg 12-hourly	CrCL < 30ml/min	400mg 24-hourly	CrCL < 25ml/min	160/800mg 12 hourly for 3 days then reduce to 160/800mg 24 hourly

Contacts	eReferral requests eReferrals Login (health.wa.gov.au)
ID registrar (phone advice) - DECT 76514	Formal ID consult referral (INTD)
ID registrar (formal ID consult request) – DECT 76515	HAPI/AMS referral (IDHA)
	Low risk penicillin allergy assessment (IDRA)